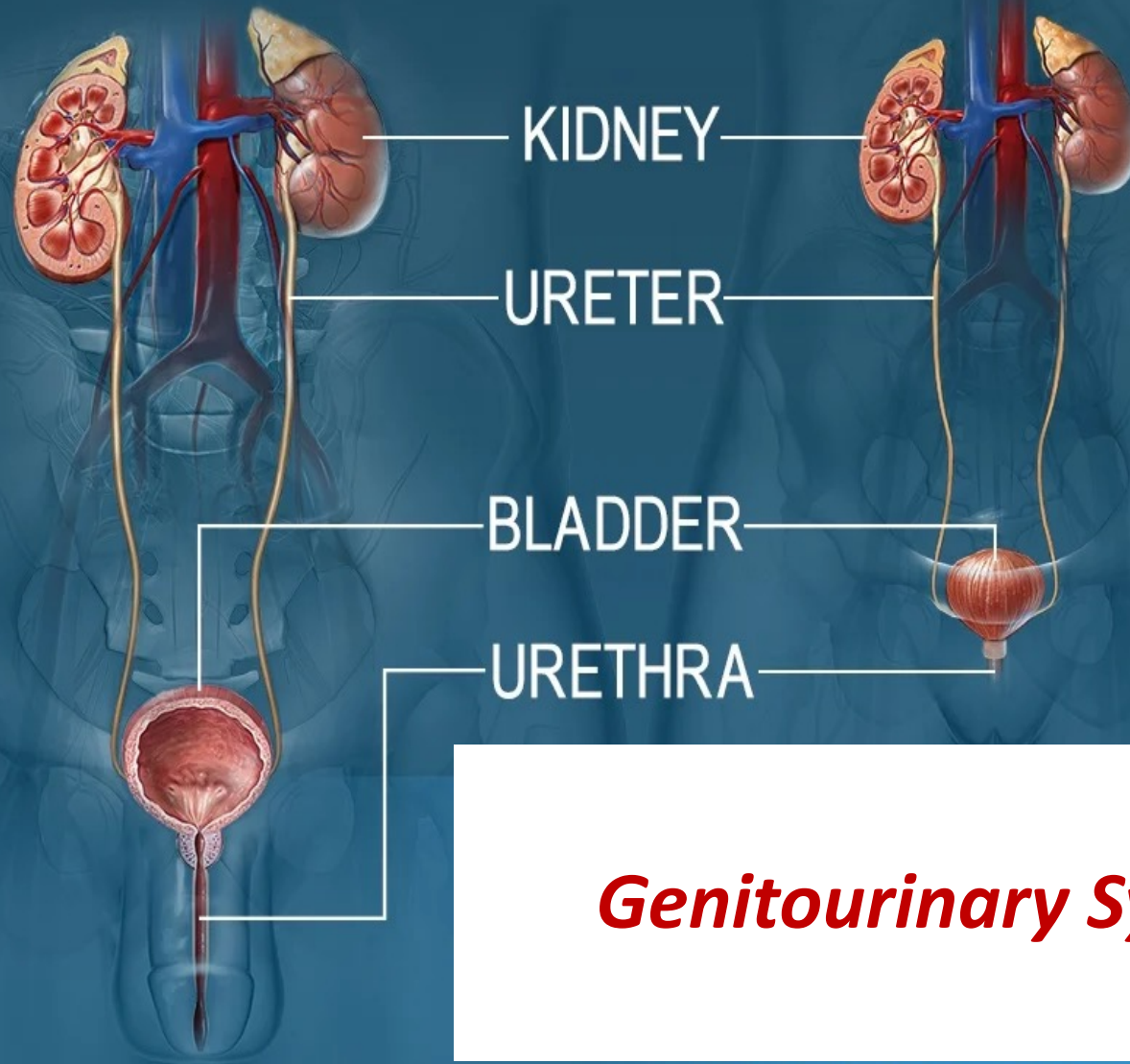


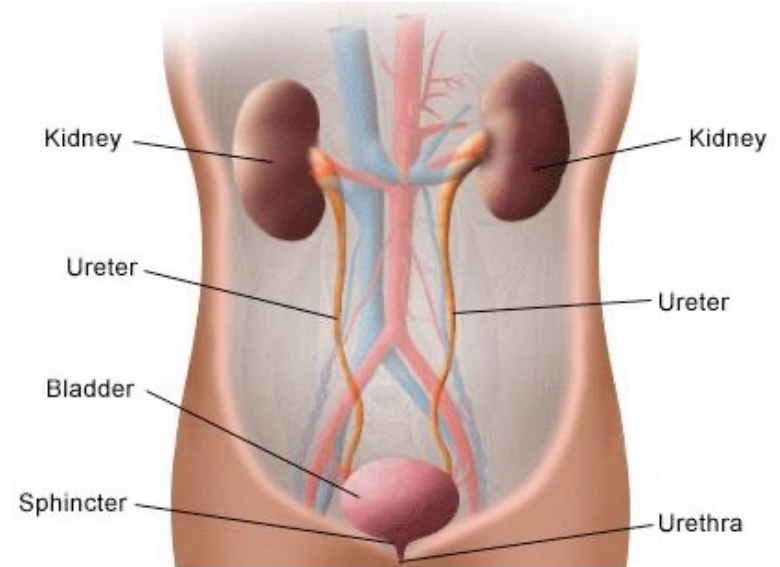


Genitourinary System

Urinary System

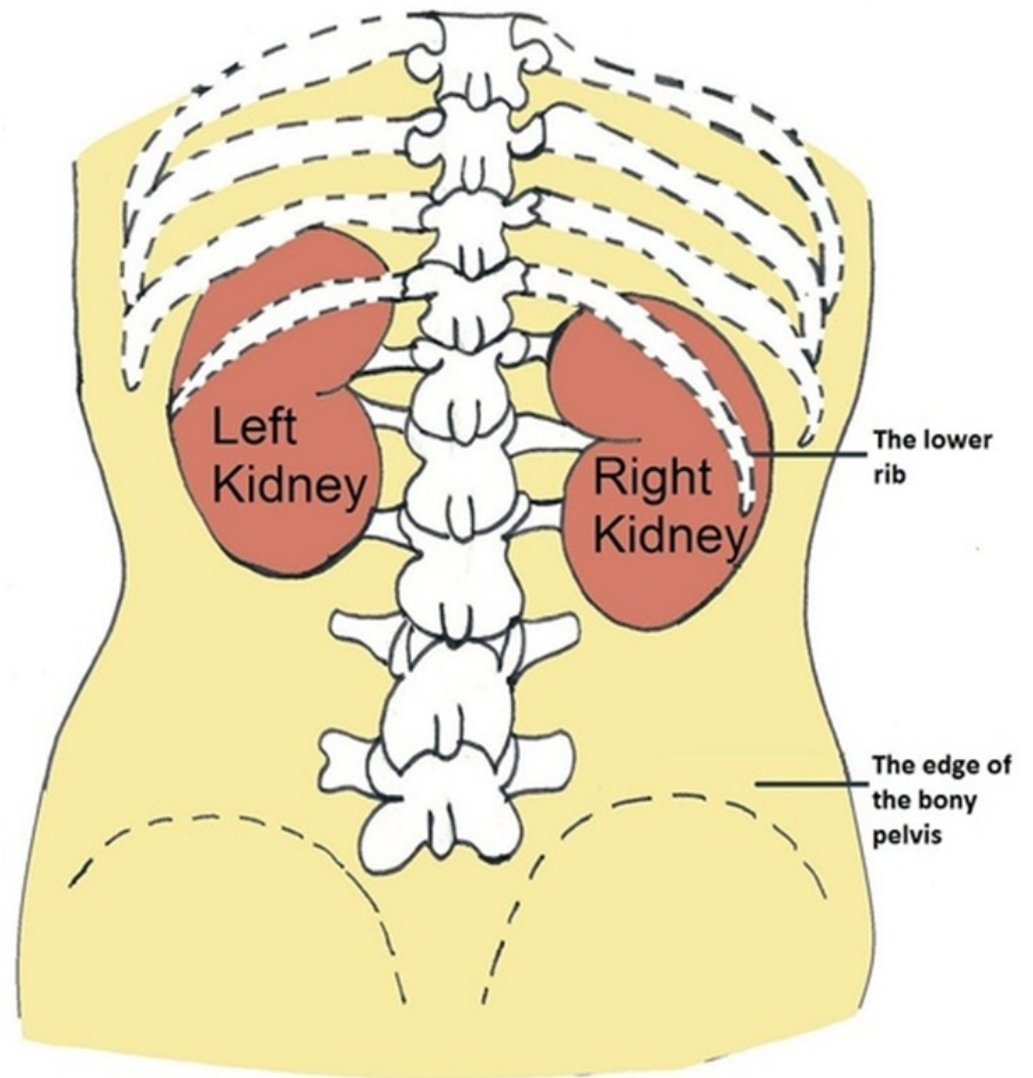
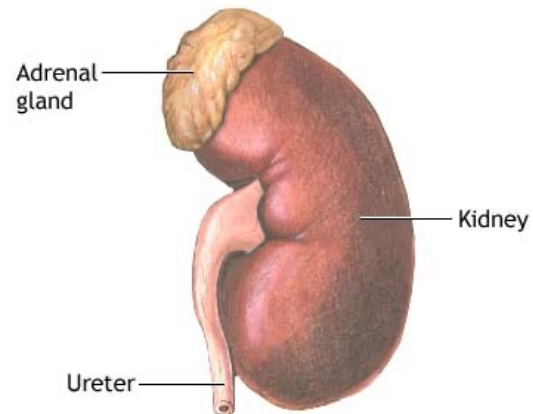
Consists of:

- ❖ Two kidneys,
- ❖ Two ureters,
- ❖ Urinary bladder &
- ❖ Urethra.



♣ The male urethra is common pathway for urine and semen as the prostatic urethra receives two ejaculatory ducts.

Kidney Position



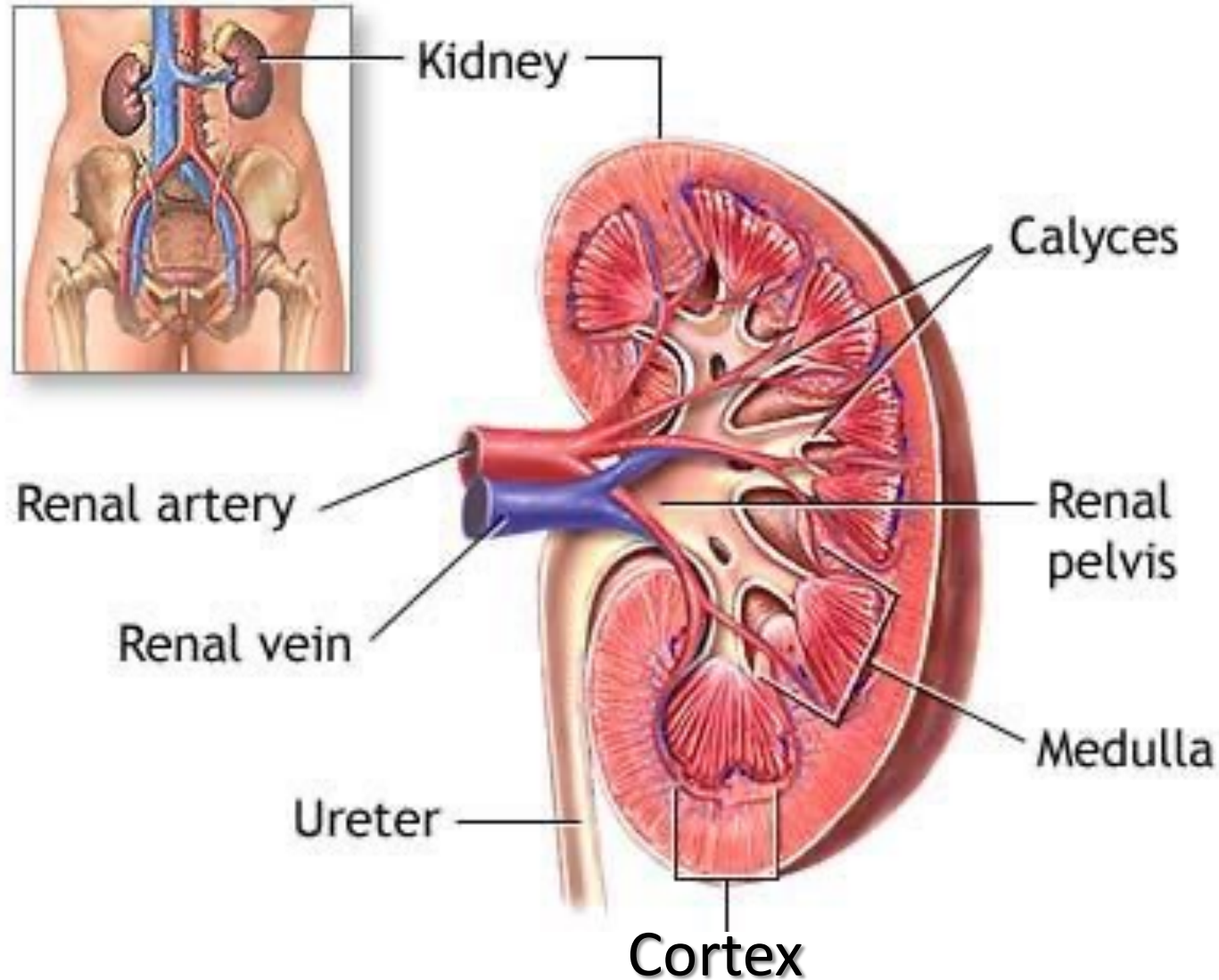
Kidneys

- The kidneys are reddish brown and lie behind the peritoneum high up on the posterior abdominal wall on either side of the vertebral column
- The right kidney lies slightly lower than the left kidney because of the large size of the right lobe of the liver.
- Each kidney has upper and lower poles, medial & lateral borders, and anterior & posterior surfaces
- On the medial concave border of each kidney is a notch called the **hilum**
- The hilum transmits the renal vein, renal artery, the ureter, and Lymph vessels
- The suprarenal or adrenal gland lies on the upper pole of each kidney

Renal structure

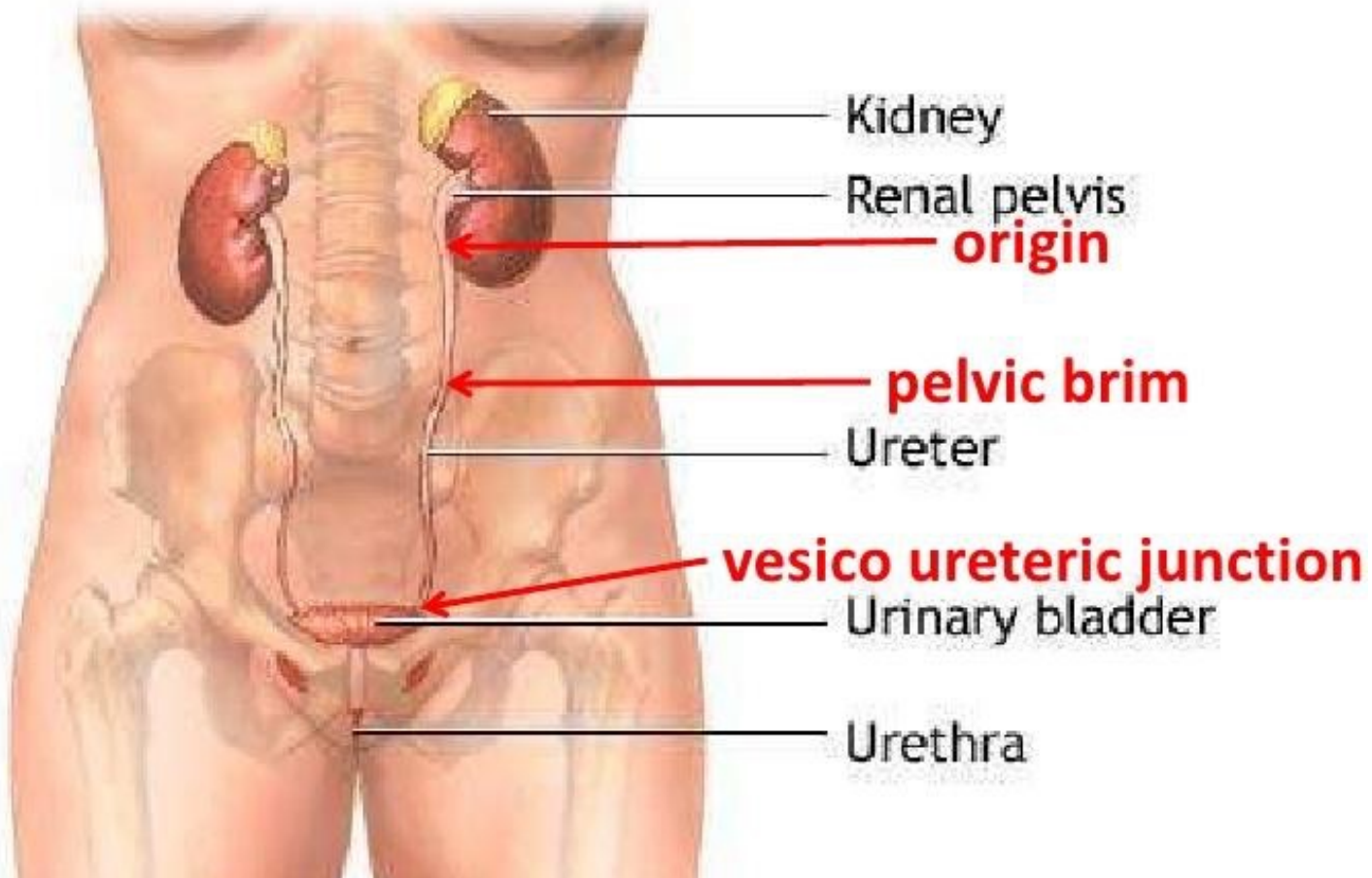
- Each kidney has a dark brown outer **cortex**, a light brown inner **medulla** and **renal sinus**.
- **The medulla** is composed of about a dozen renal pyramids.
- **The cortex** has striations known as medullary rays.
- **The renal sinus**, also called renal pelvis, contains the upper expanded part of the ureter and divides into two or three major calyces, each of which divides into two or three minor calyces

Renal structure



Ureter

CONSTRICTIONS



Ureter

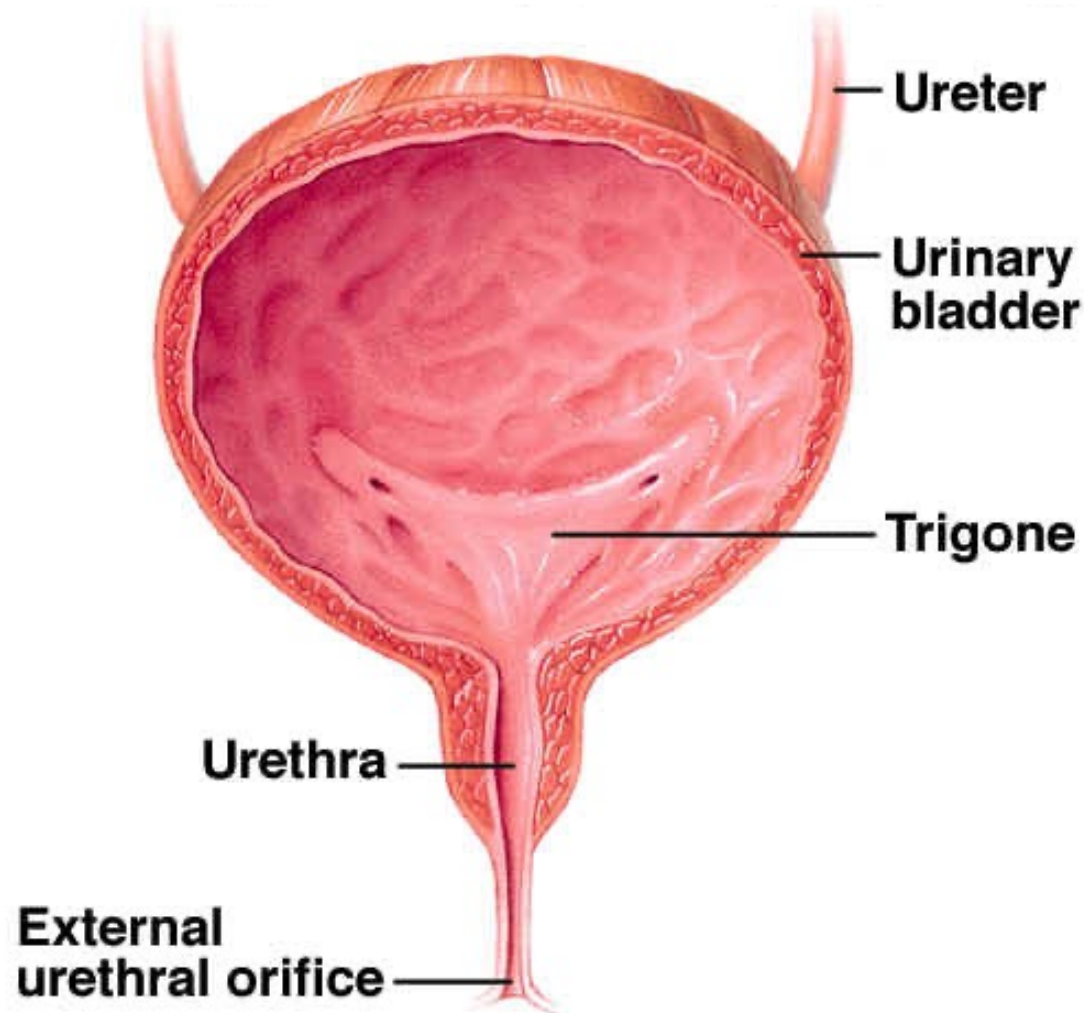
- **Location and Description**

- The two ureters are muscular tubes that extend from the kidneys to the posterior surface of the urinary bladder
- Each ureter measures about 10 in. (25 cm) long
- Ureter has three constrictions along its course:
 1. where the renal pelvis joins the ureter,
 2. where it is kinked as it crosses the pelvic brim,
 3. and where it pierces the bladder wall

Urinary Bladder:

- ♣ The bladder is a muscular sac, stores and excretes urine with a volume capacity of 500ml
- ♣ Empty bladder is completely intrapelvic, when full it ascends into the lower abdomen.
- ♣ The two ureters open into urinary bladder and the urethra starts at internal urethral orifice
- the internal surface of the base of the bladder is called the **trigone**
- The superior angles of the trigone correspond to the openings of the ureters, and the inferior angle to the internal urethral orifice

Urinary bladder



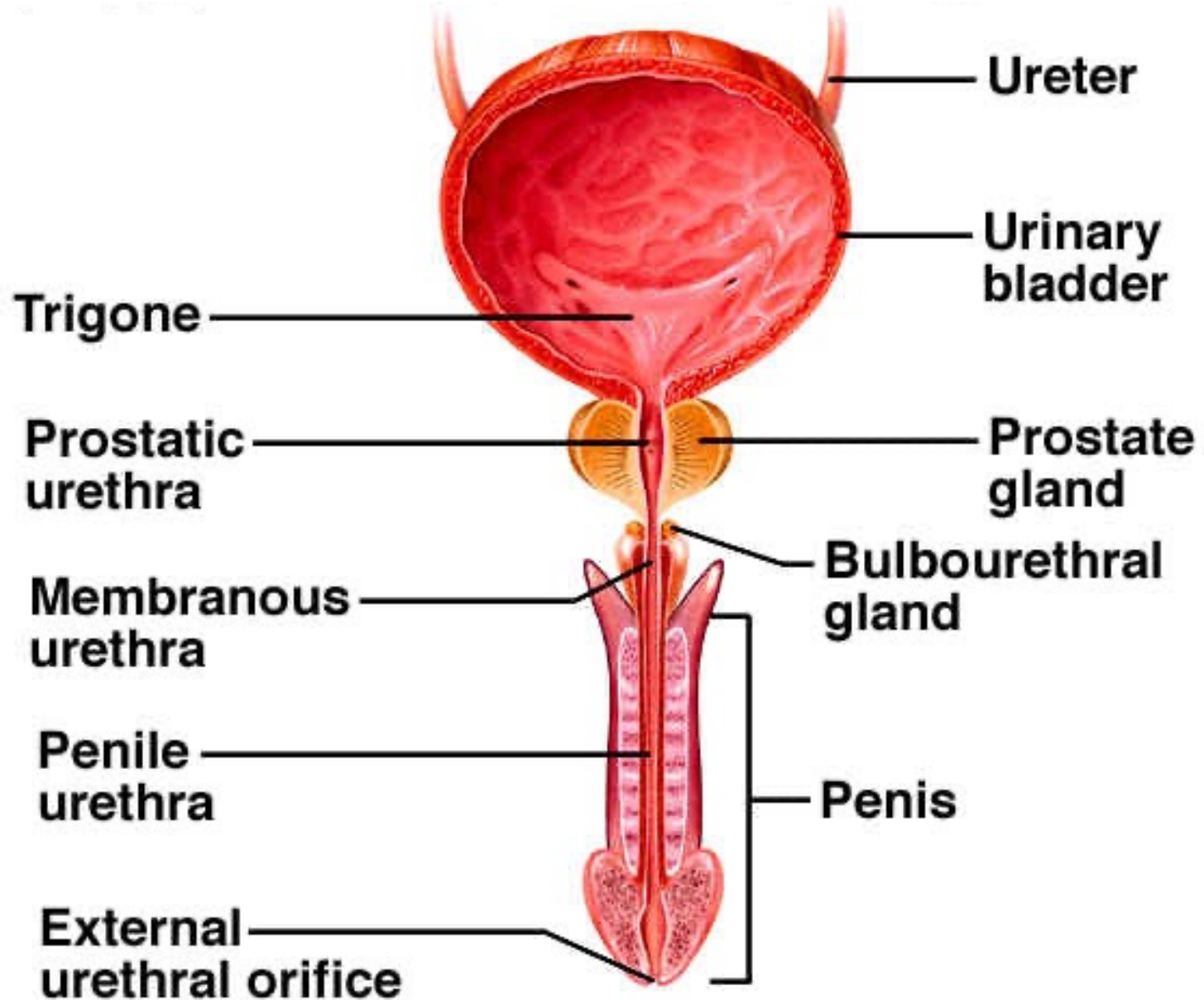
Male Urethra

- ♣ It is 20 cm long muscular tube extending from internal urethral orifice of urinary bladder to the external urethral orifice on the glans penis.
- ♣ It is divided into the *prostatic, membranous and penile parts*.
- ♣ It is common pathway for urine & semen

Female Urethra

4 cm long, begins from urinary bladder and terminates in the external urethral orifice in the vestibule anterior to the vaginal opening.

Male Urinary bladder & Urethra



Male Reproductive System

Consists of:

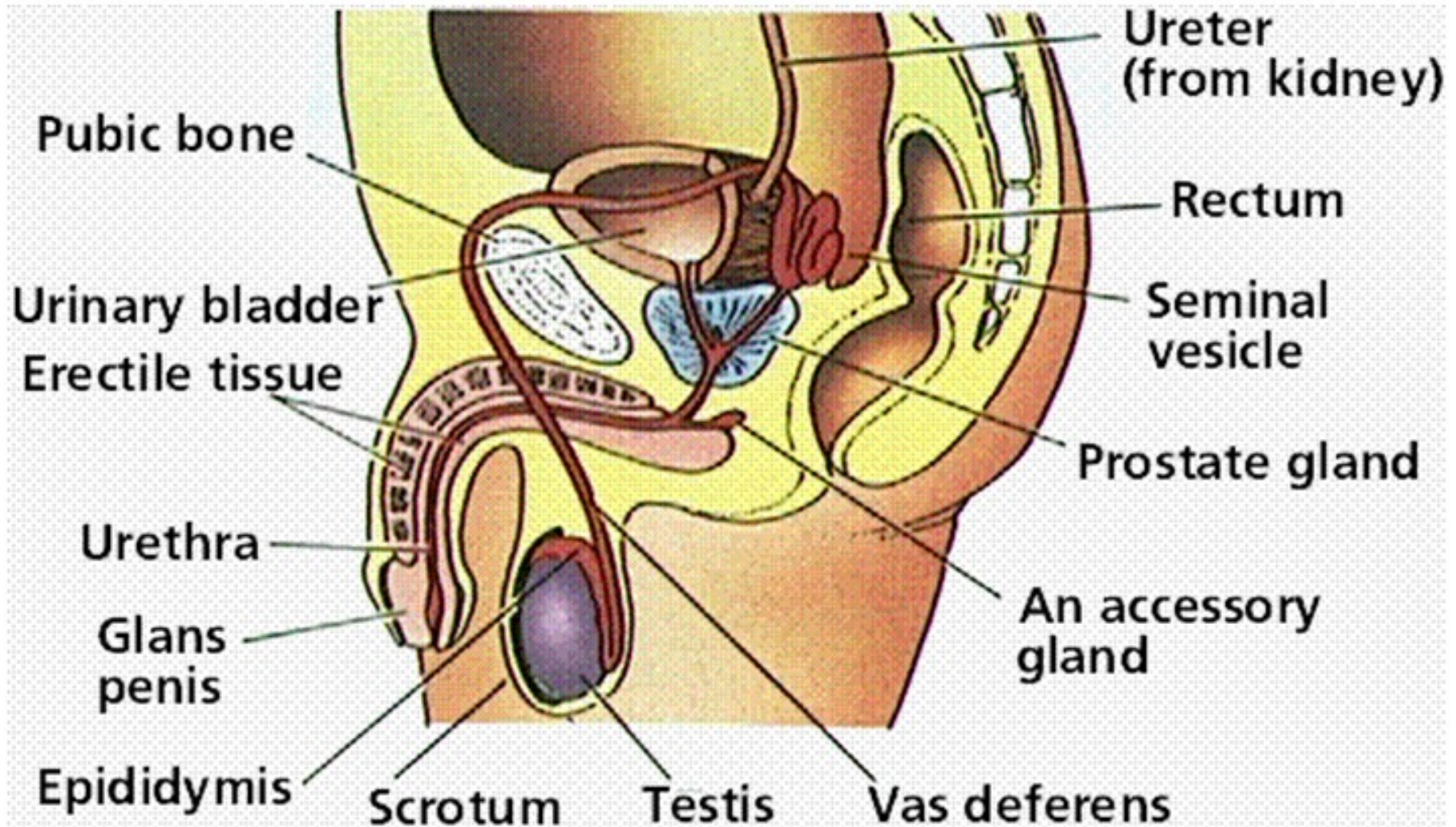
1- Testis

2- Ducts: Epididymis,
Ductus or vas deferens
Ejaculatory duct,
Urethra

3- Accessory sex gland: seminal vesicles
prostate gland
bulbourethral glands

4- External genitalia: penis scrotum

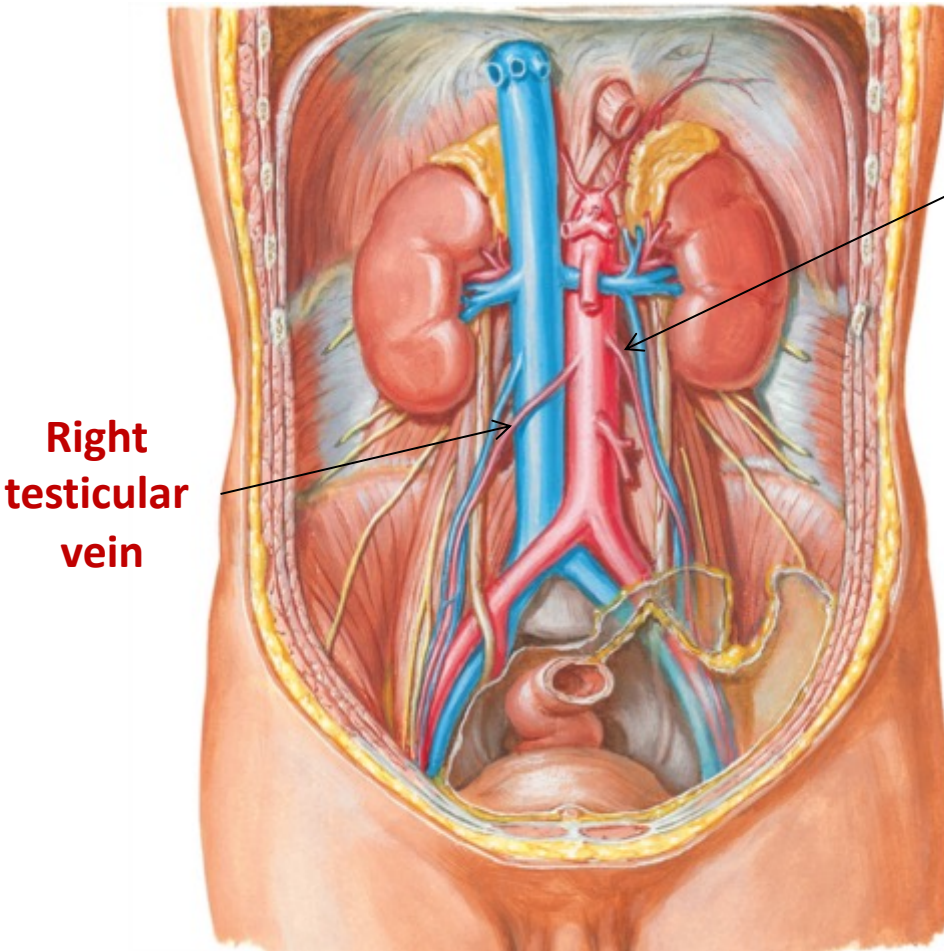
Male Reproductive System



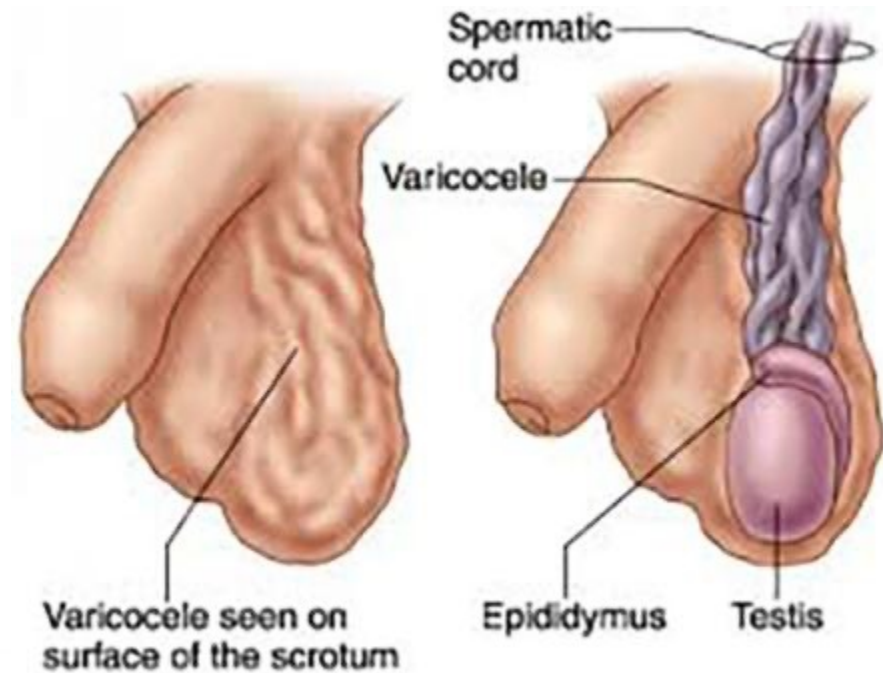
Testes

- Each testis is an oval structure about 5 cm long and 3 cm in diameter
- Located in the scrotum and surrounded by tunica albuginea
- the **tunica albuginea** extending from the inner surface that divide the interior of the organ into **lobules**
- There are many lobules in each testis. Each lobule contains 1 to 4 - **seminiferous tubules**.
- The tubules open into a network of channels called the **rete testis**. **Small efferent ductules connect** the rete testis to the upper end of the epididymis .
- Interstitial cells (cells of Leydig), which produce male sex hormones testosterone , are located between the seminiferous tubules within a lobule.

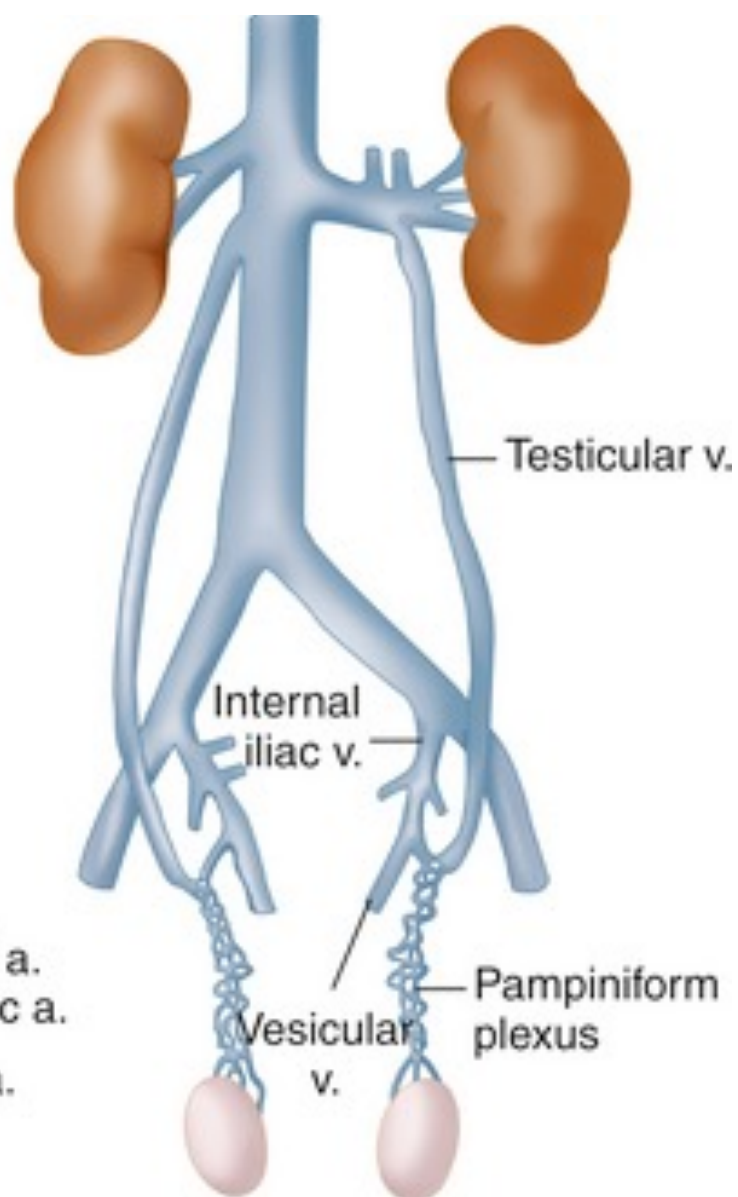
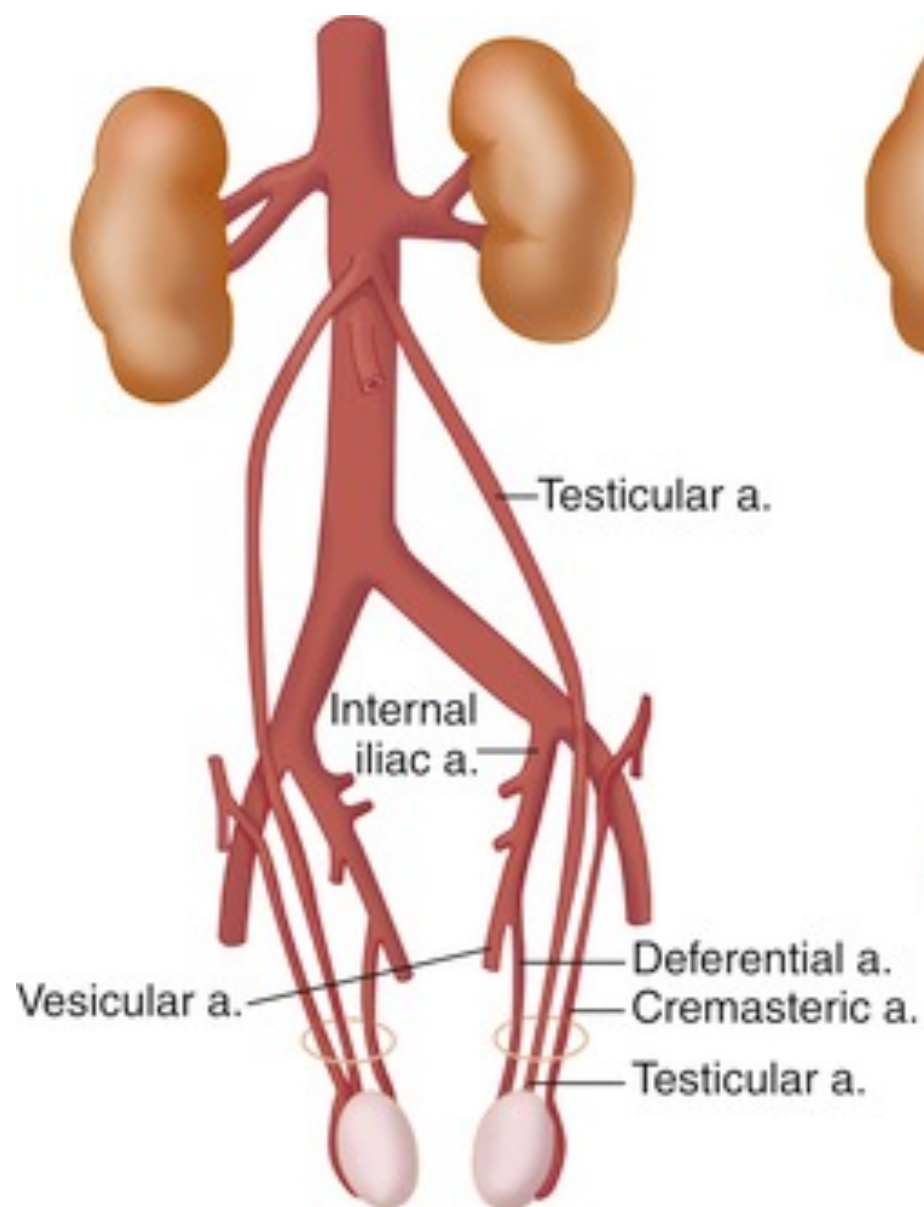
Blood vessels of the Tests



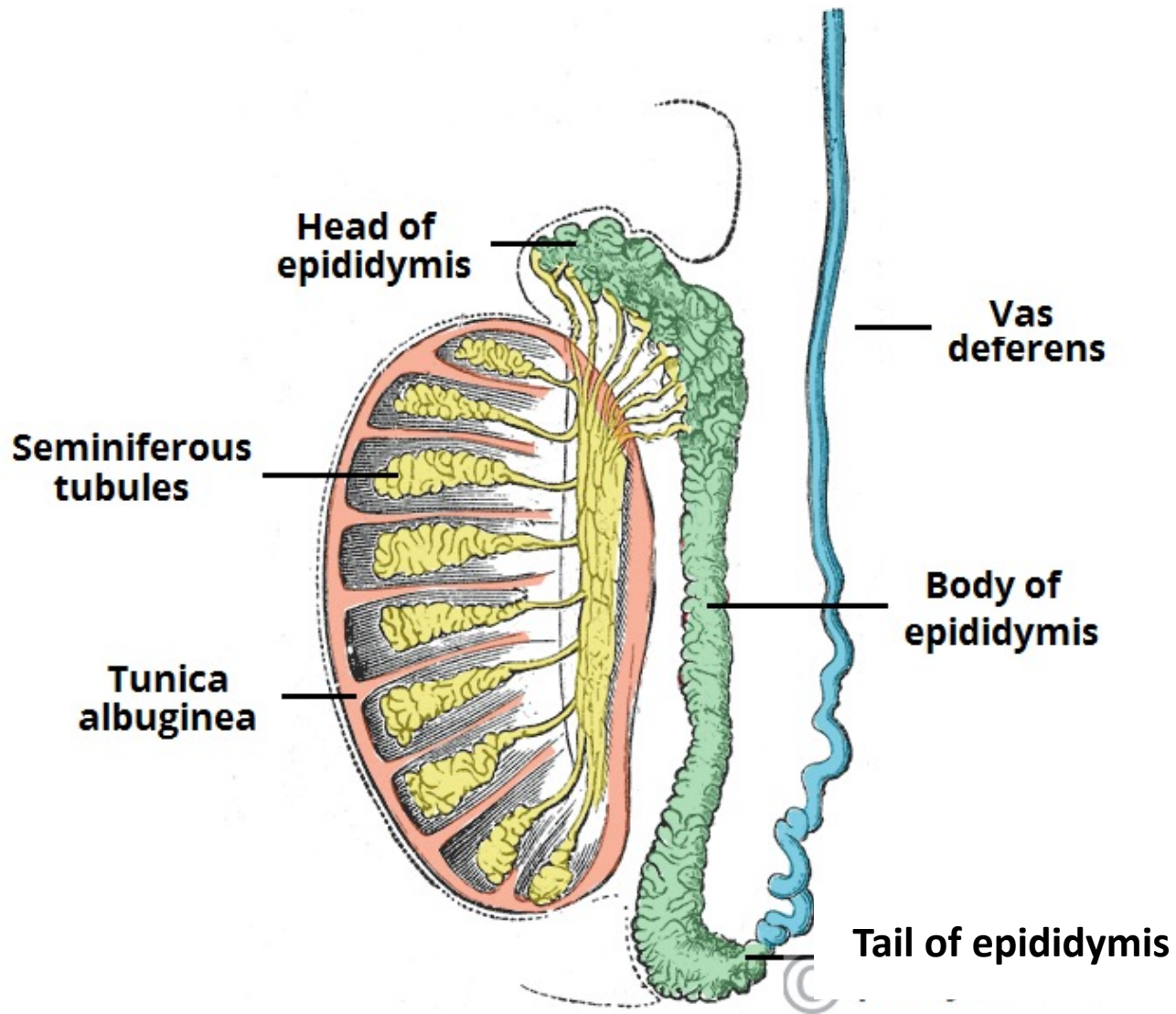
Left testicular vein



Varicocele



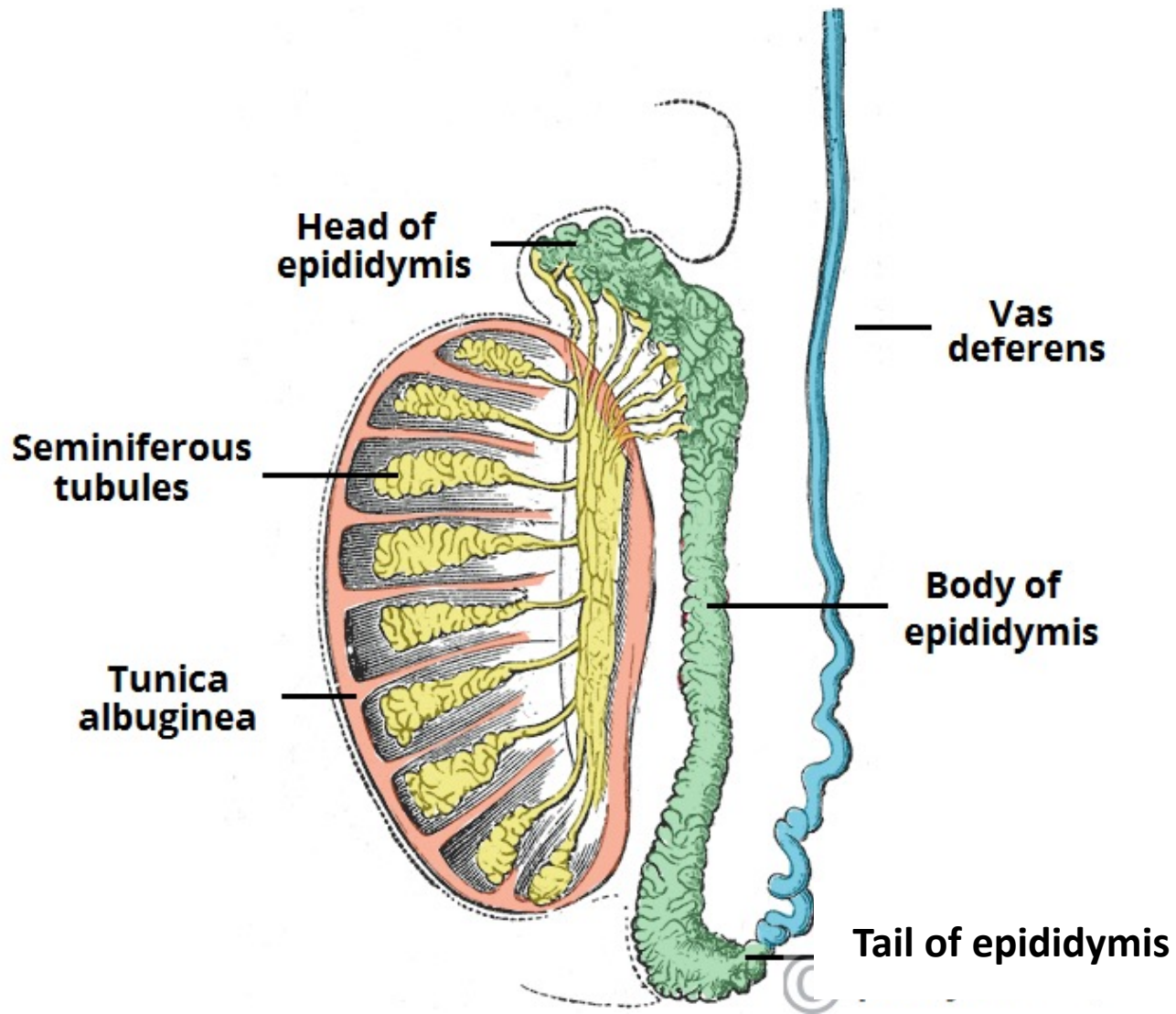
Testes



Epididymis

- The epididymis is a coiled tube located posterior to the testes and is about 6 meters
- It has an expanded upper end, *the head, a body, and tail*.
- The tube emerges from the tail of the epididymis as the *vas deferens*, which enters the spermatic cord.
- The long length of the duct of the epididymis provides storage space for the spermatozoa and allows them to mature.

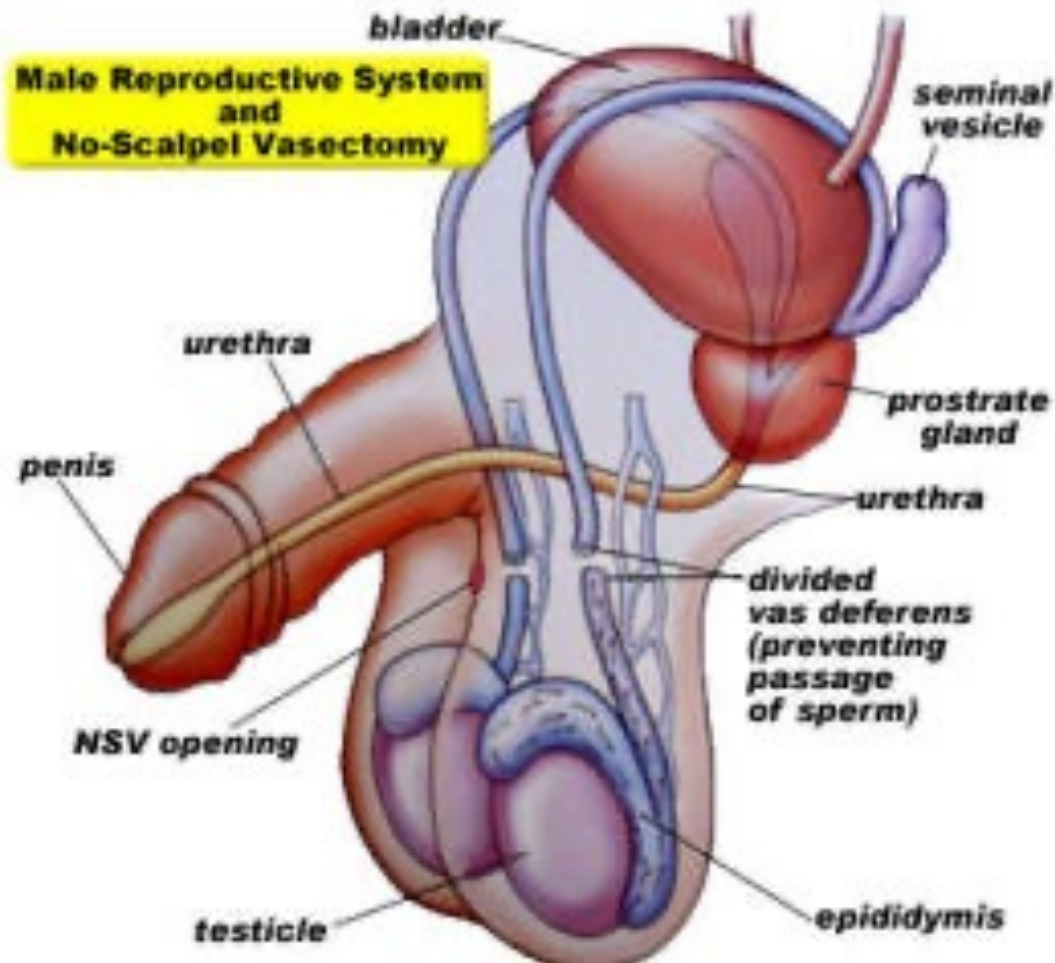
Epididymis



Vas Deferens

- The vas deferens is a thick-walled tube about (45 cm) long that conveys mature sperm from the epididymis to the ejaculatory duct and the urethra.
- It arises from the lower end or **tail of the epididymis**.
- The vas deferens enter to the abdomen through the inguinal canal then run backward on the **posterior surface of the bladder**.
- The terminal part of the vas deferens is dilated to form the **ampulla of the vas deferens** and joins with the duct of the seminal vesicle to form the **ejaculatory duct**.

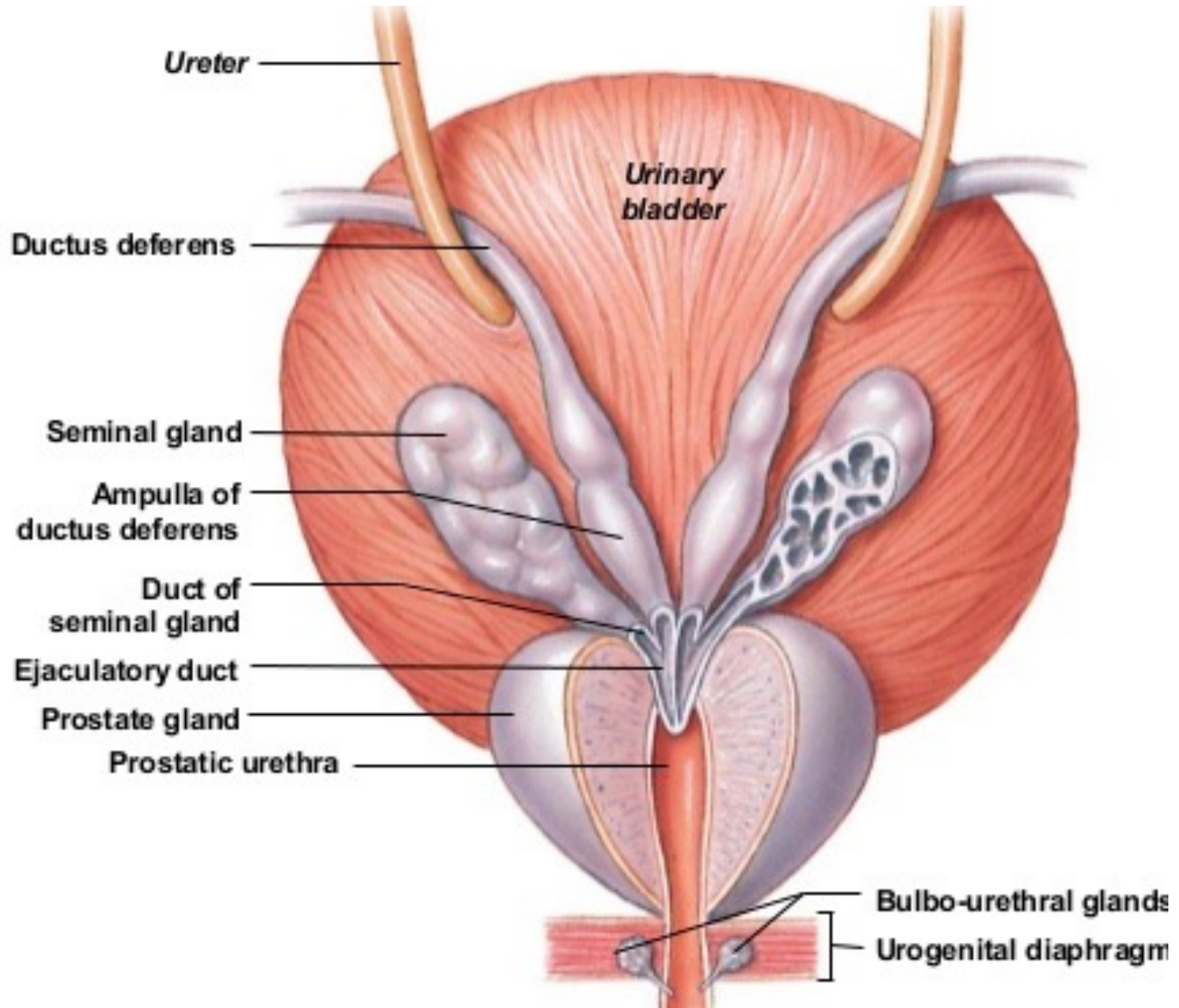
Vas deferens



The seminal vesicles

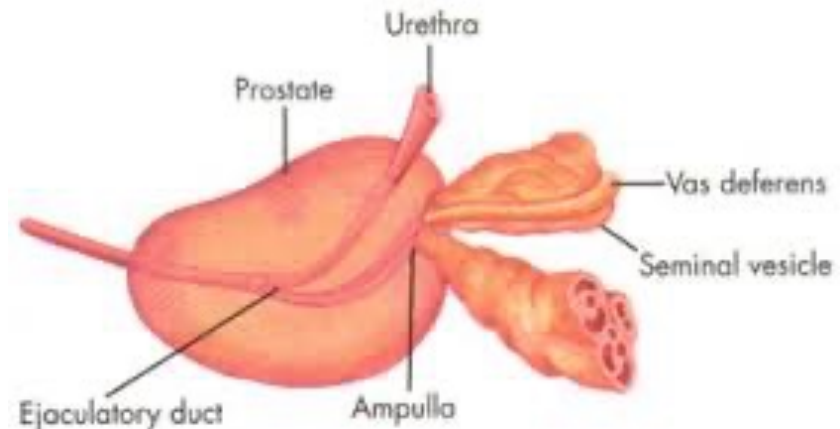
- The seminal vesicles are two lobulated organs lying on the posterior surface of the bladder
- Each terminal part of vesicle joins with the vas deferens to form the ejaculatory duct.
- **Function**
- The function of the seminal vesicles is to produce a secretion that is added to the seminal fluid.

Seminal vesicle



Ejaculatory Ducts

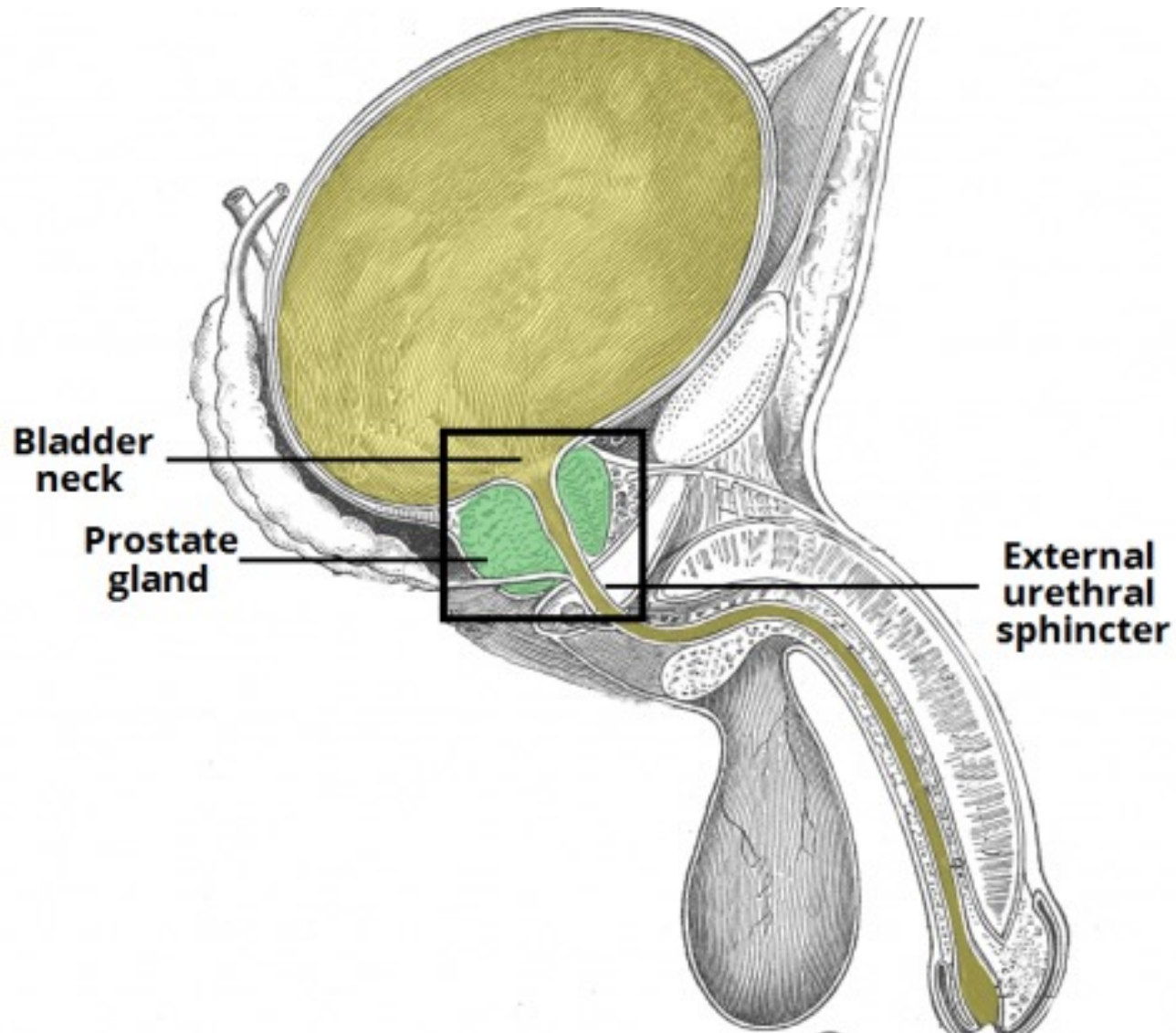
- The two ejaculatory ducts are (2.5 cm) long and are formed by the *union of the vas deferens and the duct of the seminal vesicle*.
- The ejaculatory ducts pierce the **prostate** and open into the prostatic part of the urethra.
- their function is to drain the seminal fluid into the prostatic urethra.



Prostate

- The prostate is a fibromuscular glandular organ that surrounds the **prostatic urethra**.
- It is about 1.25 in. (3 cm) long and lies between the neck of the bladder
- The two ejaculatory ducts pierce the prostate
- The secretions of the prostate are thin, milky colored, and alkaline. They function to enhance the motility of the sperm.

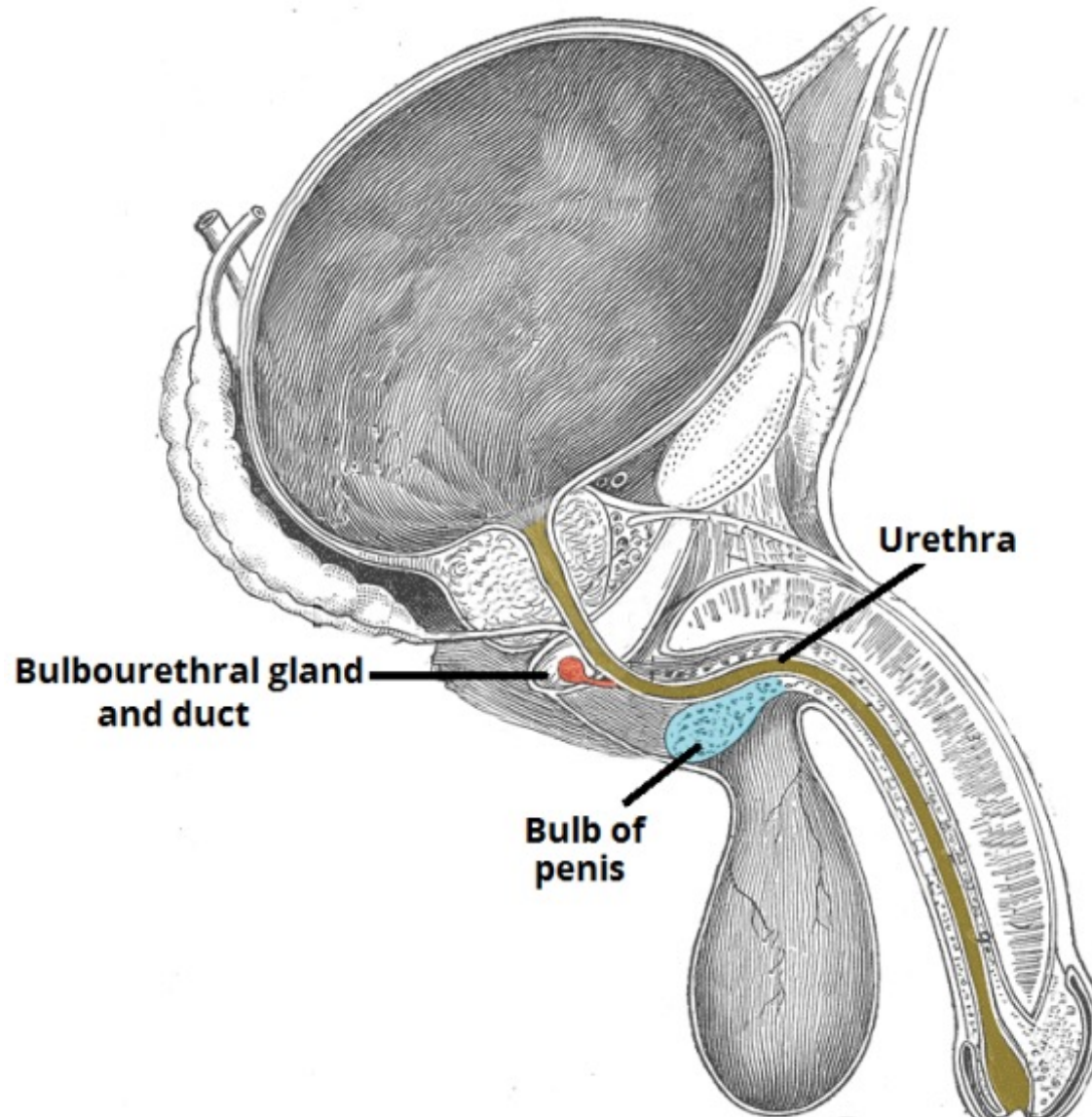
Prostate gland



Bulbourethral Glands (Cowper's)

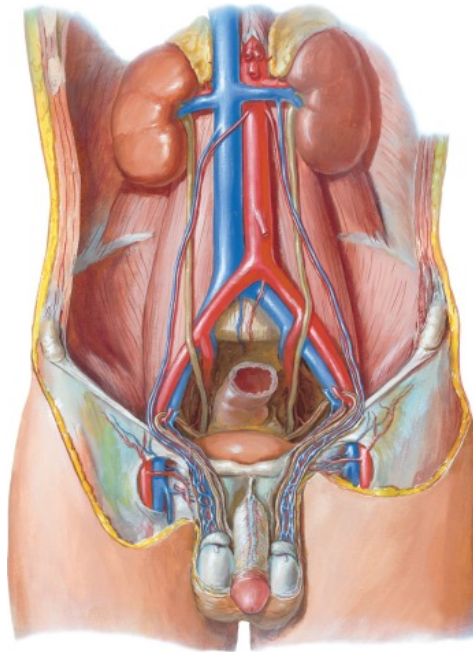
- small, about the size of a pea, and located near the base of the penis.
- Its duct opens in to the proximal end of the penile urethra.
- In response to sexual stimulation, also secrete an alkaline mucus-like fluid

Bulbourethral gland



Scrotum

- The scrotum is an outpouching of the lower part of the anterior abdominal wall separated by raphe and contains the testes, the epididymides, and the lower ends of the spermatic cords

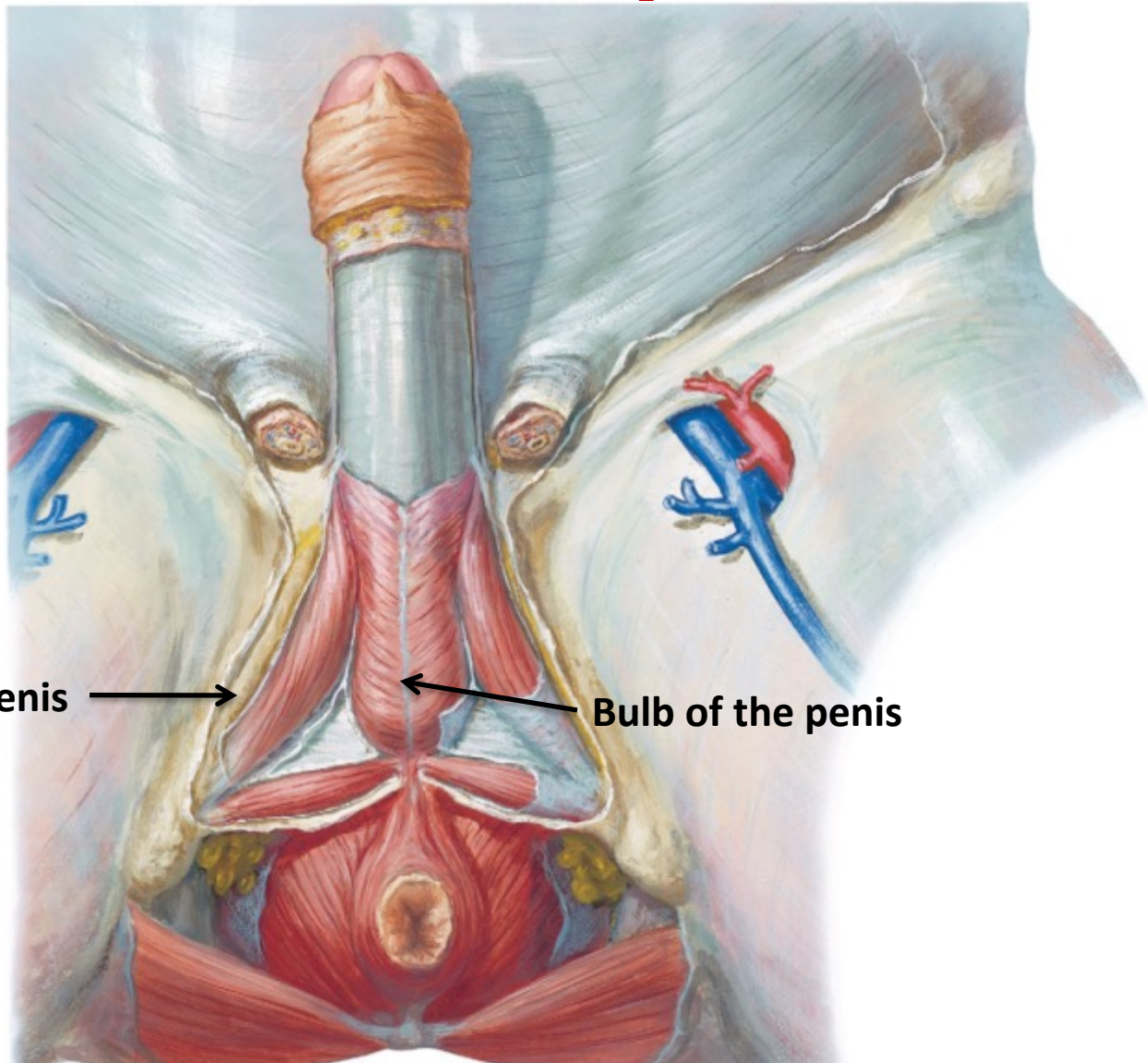


Penis

- The penis has a fixed root and a body that hangs free
- Root of the Penis
- The root of the penis is made up of three masses of erectile tissue called the **bulb of the penis and the right and left crura of the penis**

Root of the penis

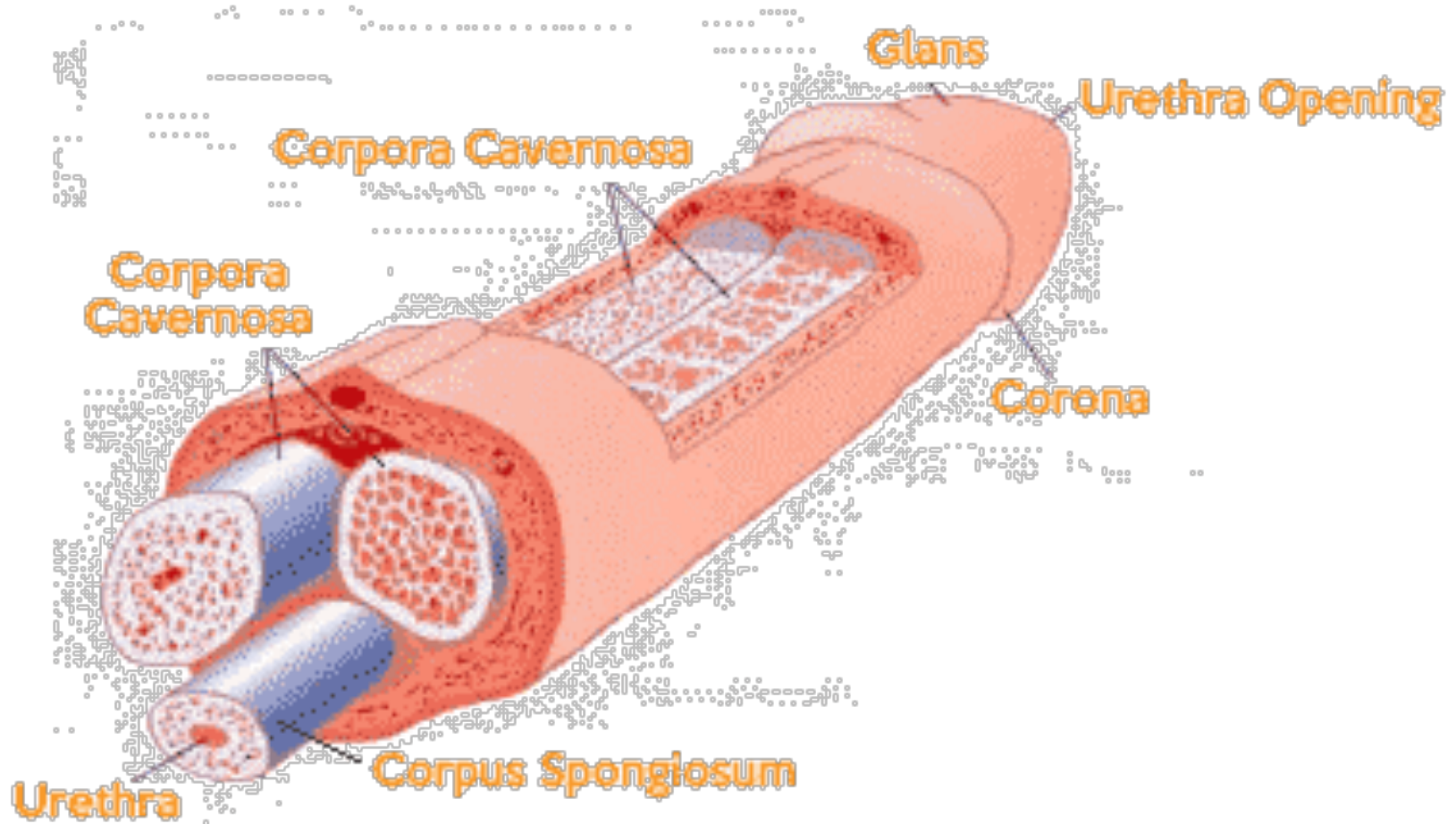
Crura of the penis → ← Bulb of the penis



Body of the Penis

- The body of the penis is essentially composed of three cylinders of erectile tissue enclosed in a tubular sheath of fascia (**Buck's fascia**).
- **The erectile tissue is made up of** two dorsally placed corpora cavernosa and a single corpus spongiosum applied to their ventral surface. At its distal extremity, the corpus spongiosum expands to form the **glans penis**
- On the tip of the glans penis is the orifice of the urethra, called the **external urethral meatus**

Body of the penis



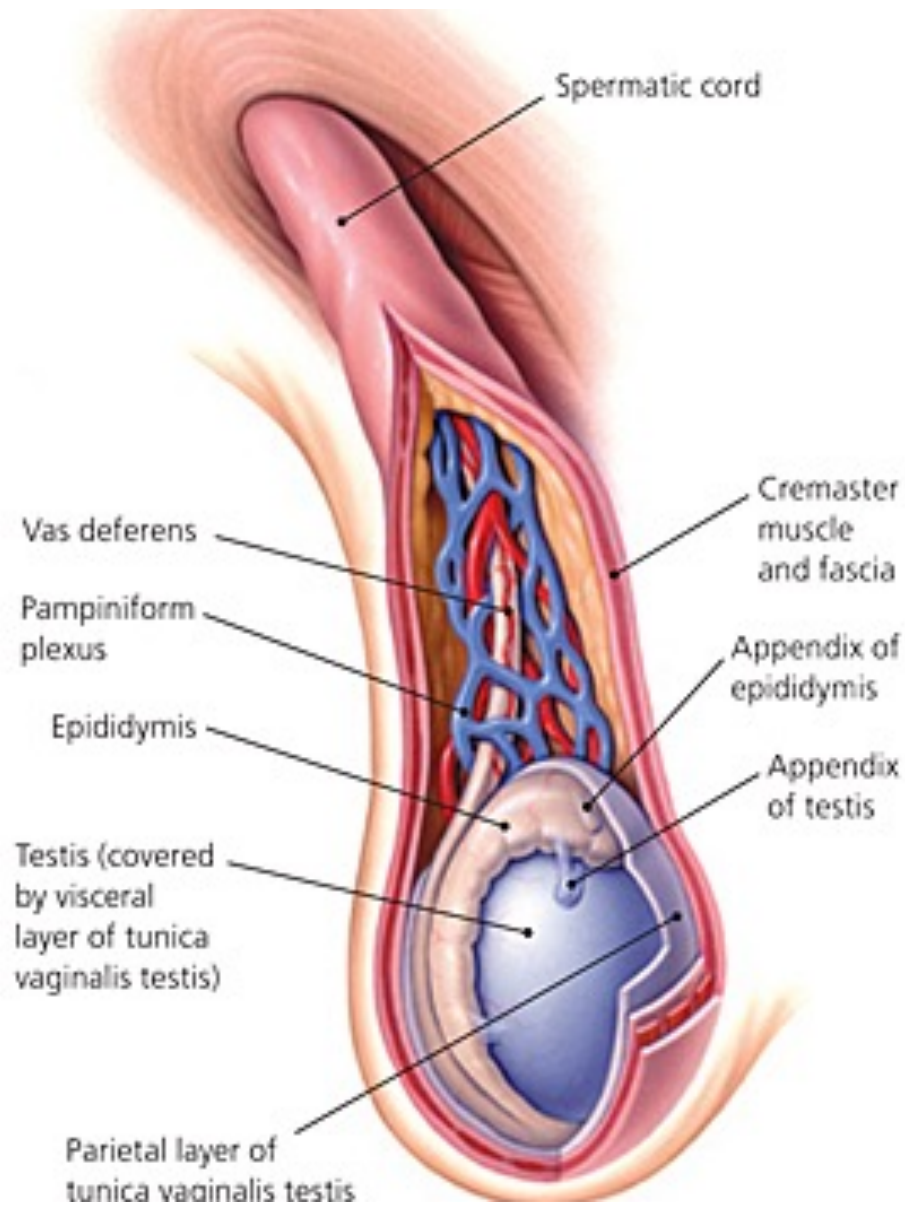
Spermatic Cord

- The spermatic cord is a collection of structures that pass through the inguinal canal to and from the testis
- **Structures of the Spermatic Cord**

The structures are as follows:

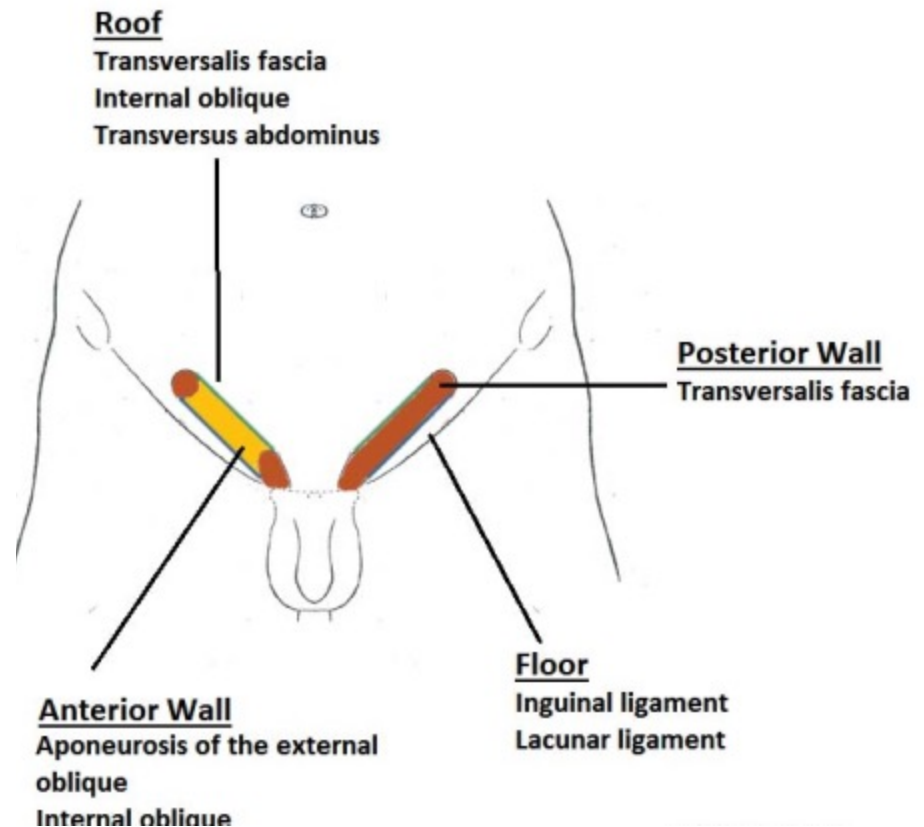
- **Vas deferens**
- **Testicular artery**
- **Testicular veins (pampiniform plexus)**
- **Testicular lymph vessels**
- **Autonomic nerves**
- **Genital branch of the genitofemoral nerve**

Spermatic cord



Inguinal canal

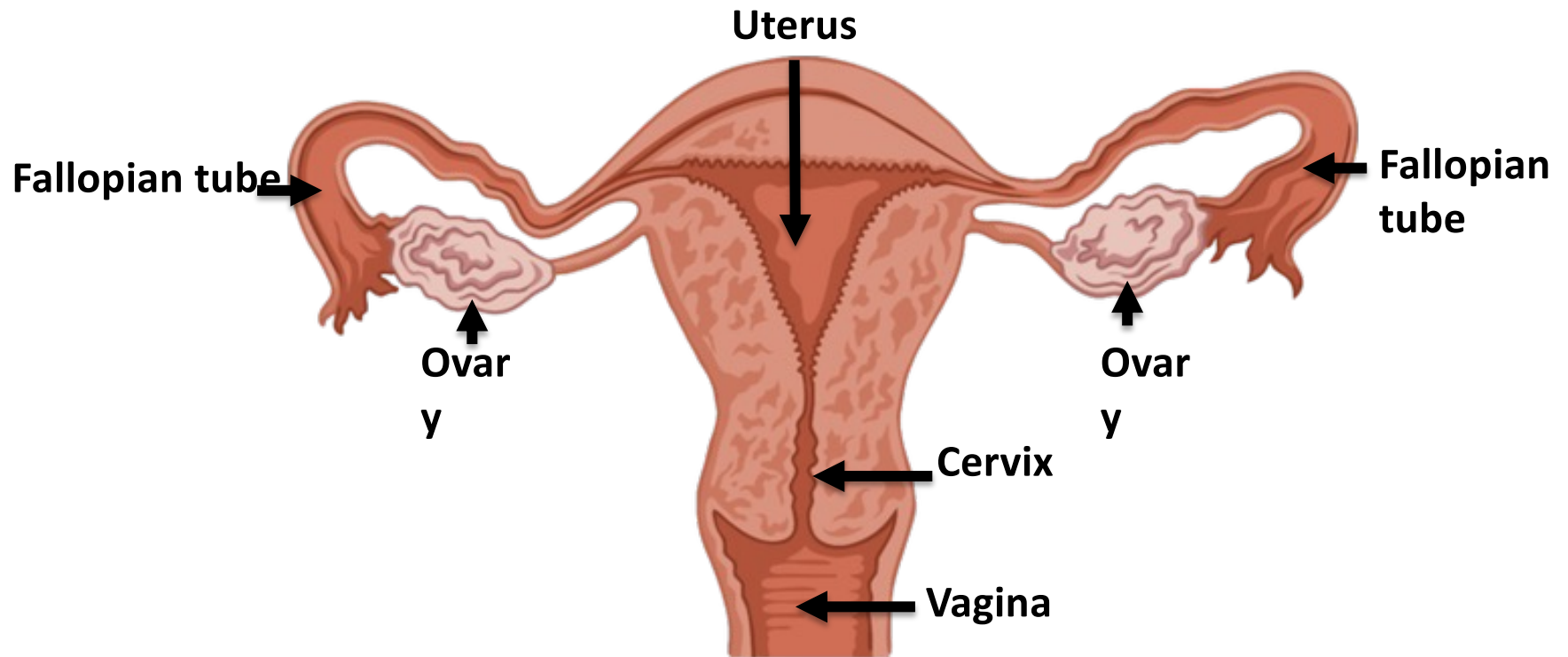
- The inguinal canal is a short passage that extends inferiorly and medially, through the **inferior part of the abdominal wall**.
- It is superior and parallel to the **inguinal ligament**.



Female Reproduction System

- The female reproductive system consists of the **external genital organs and internal genital organs.**
- I. **Internal Genital Organs including:**
 - ovaries, uterine or fallopian tubes, uterus and vagina.
- II. **External genital organs called (Vulva) and consist of:**
 - labia majora, labia minora, vestibule of the vagina, bartholin's glands and clitoris

Internal genital organs



Ovaries

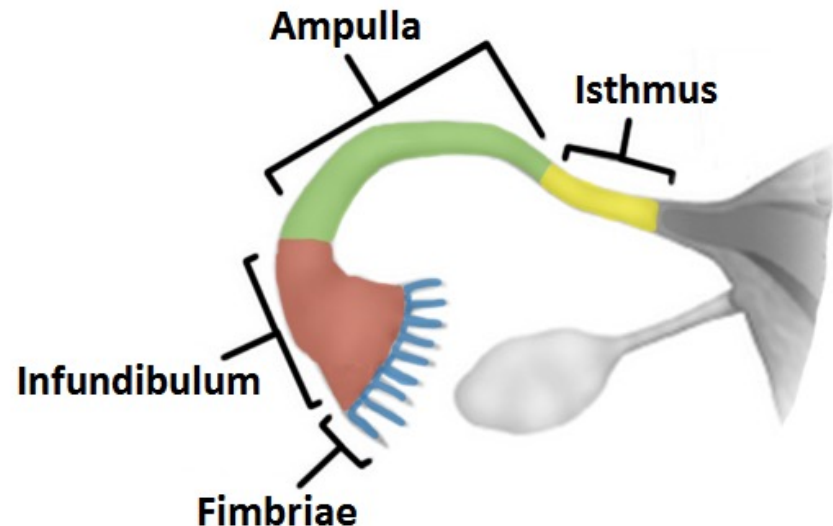
- Located in the **upper pelvic cavity**, on either **side of the uterus**.
- They are maintained in position by ligaments (**suspensory ligament, ovarian ligament and broad ligament**).
- The ovary consists of large number of **primary follicles**.
- The main functions of ovaries are **producing the ovum, estrogen and progesterone**.

Uterine tubes

- **Extend from the uterus to the ovaries** and connect the uterine cavity to the peritoneal cavity.
- Are each subdivided into four parts:
 1. **The uterine part (intra mural part) ,**
 2. **The isthmus ,**
 3. **The ampulla (the longest and widest part),**
 4. **And the infundibulum (the funnel-shaped termination formed of fimbriae).**

Uterine tubes

- **Convey the fertilized eggs** to the uterus by ciliary action and muscular contraction
- **Fertilization takes place** within the tube, usually **in the infundibulum or ampulla**.



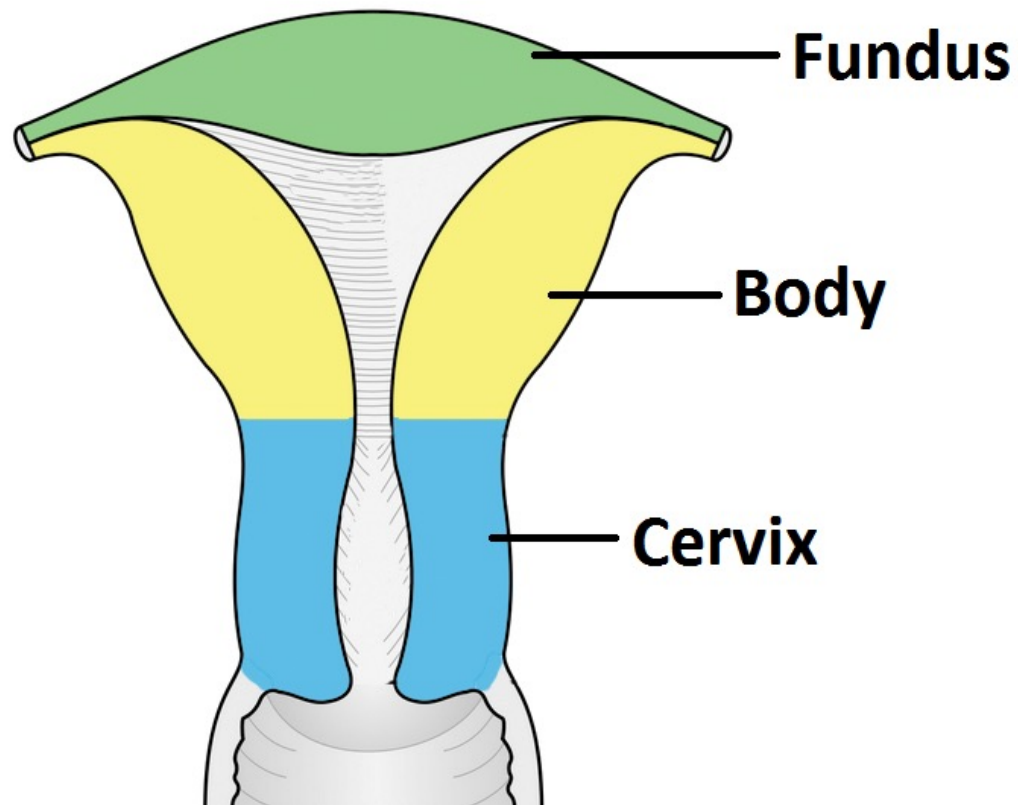
Uterus

- Is the organ of gestation in which the fertilized oocyte normally becomes embedded and the developing organism grows until its birth.
- Is divided into four parts for the purpose of description:
 1. **Fundus:** Is the **rounded part of the uterus and receives the entrance of** the uterine tube.
 2. **Body:** Is the main part of the uterus located inferior to the fundus and superior to the isthmus.
- The uterine cavity is triangular shape.

Uterus

3. **Isthmus:** Is the **constricted part of the uterus located between the body and cervix of the uterus.**
 - It corresponds to the internal os.
4. **Cervix:** Is the inferior narrow part of the uterus that projects into the vagina and divides into the following regions:
 - **Internal os: the junction of the cervical canal with the uterine body.**
 - **Cervical canal : the cavity of the cervix between the internal and external ostia.**

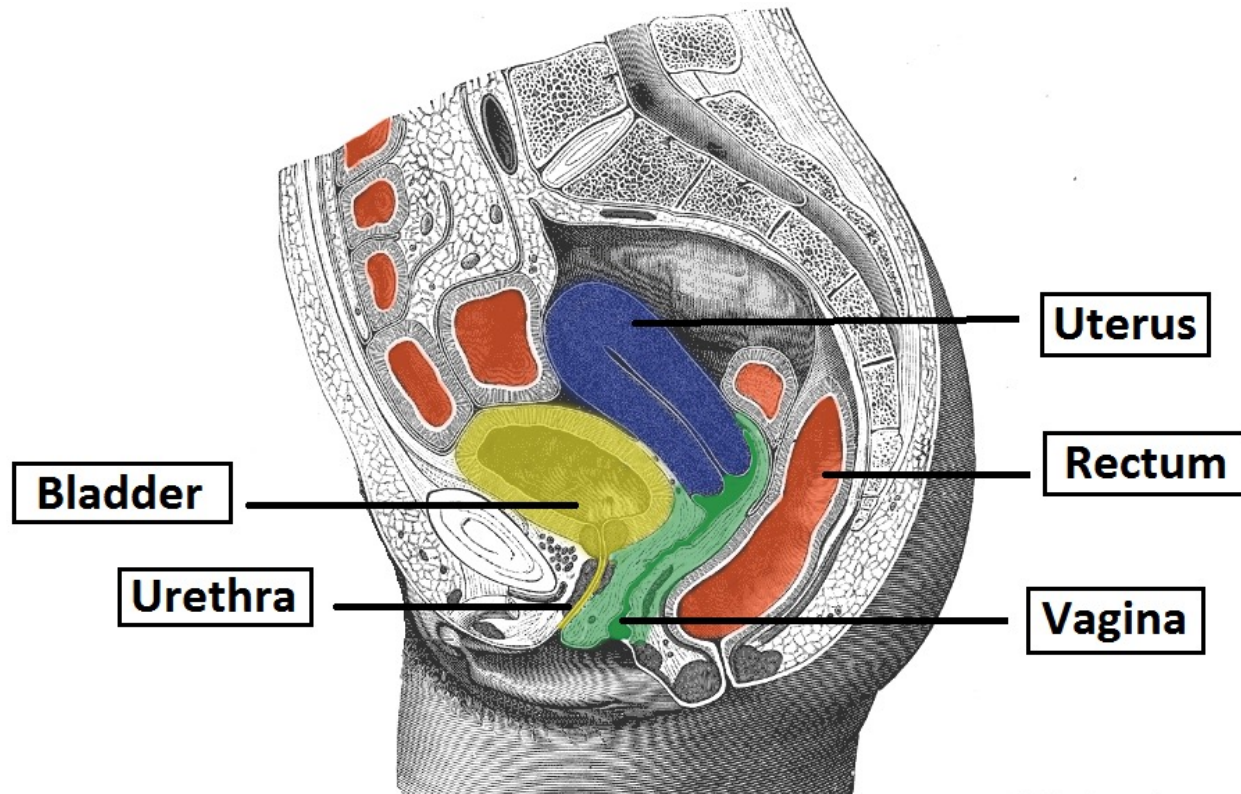
Parts of the uterus

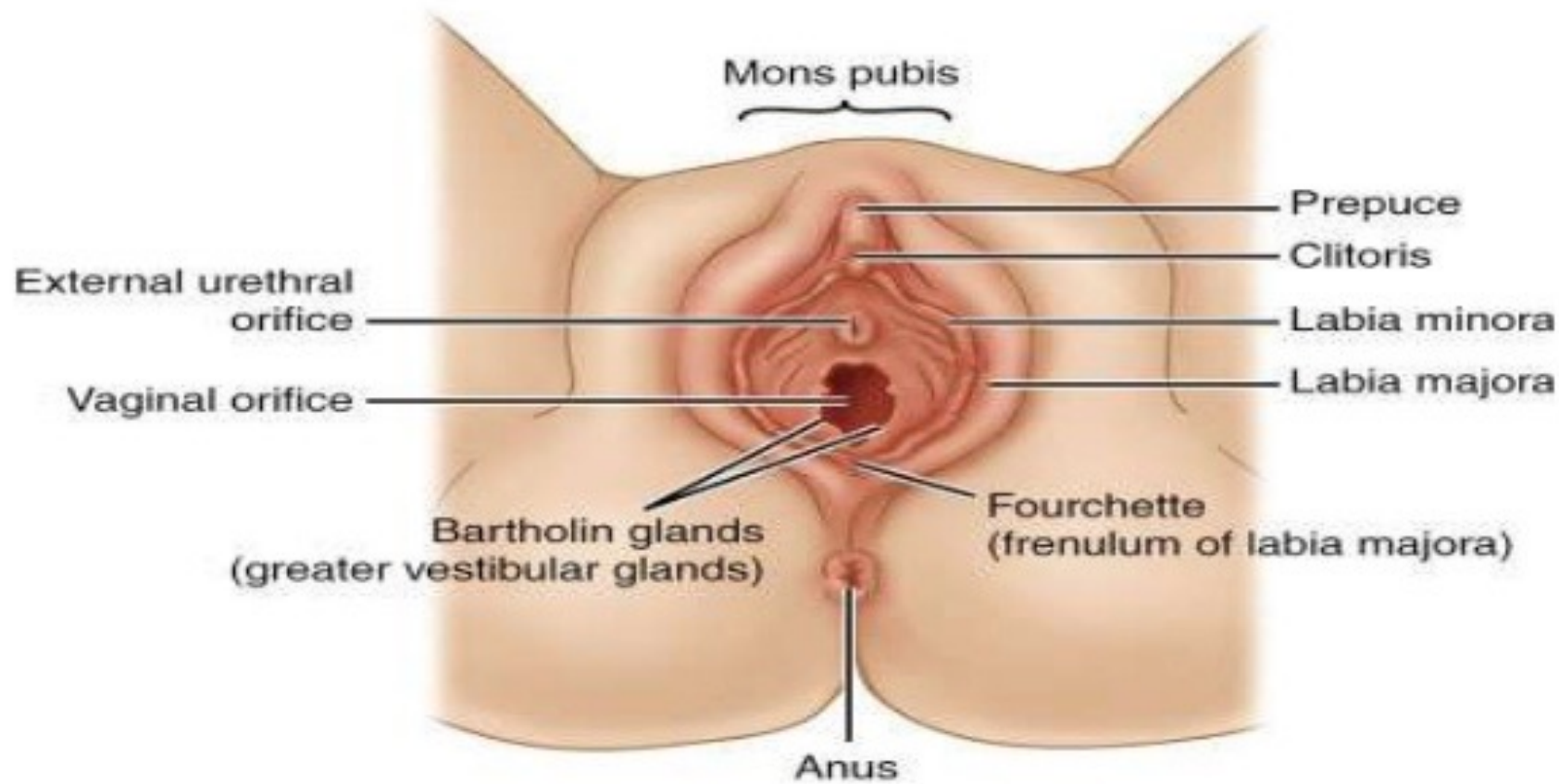


Vagina

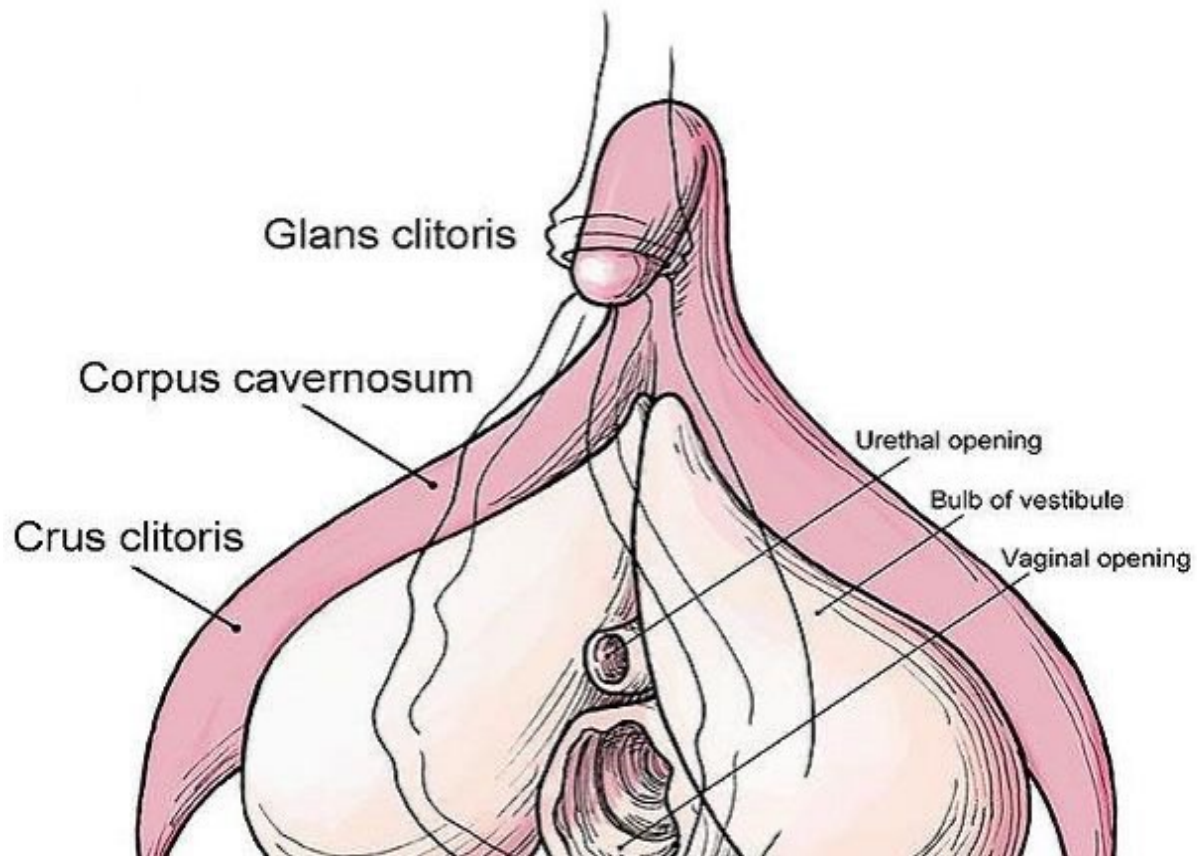
- The vagina is a muscular tube about (10 cm) long that **extends from the cervix to the vaginal orifice in the perineum.**
- The vagina locates **between the bladder anteriorly and rectum posteriorly.**
- It lined by multiple folds called rugae, these folds allow the vagina to stretch during childbirth.
- **The main function of the vagina are**
 1. Allow discharge of the menstrual flow.
 2. As the female organs of coitus.
 3. Allow passage of fetus from the uterus.

Vagina

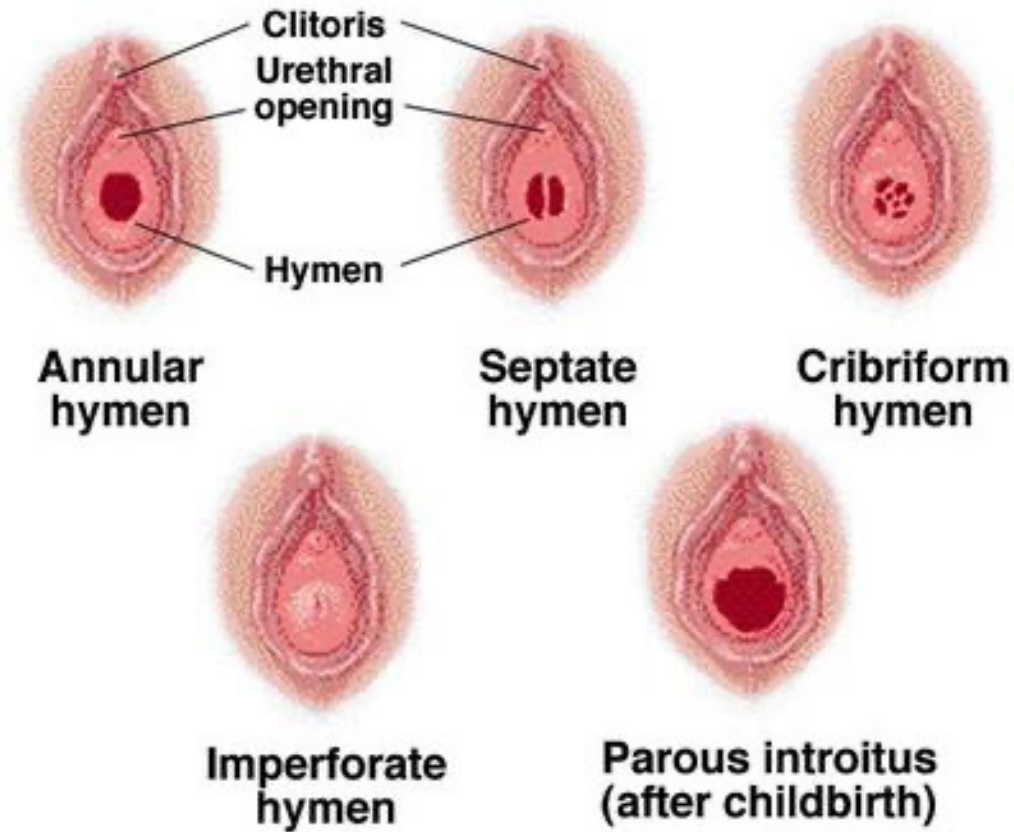




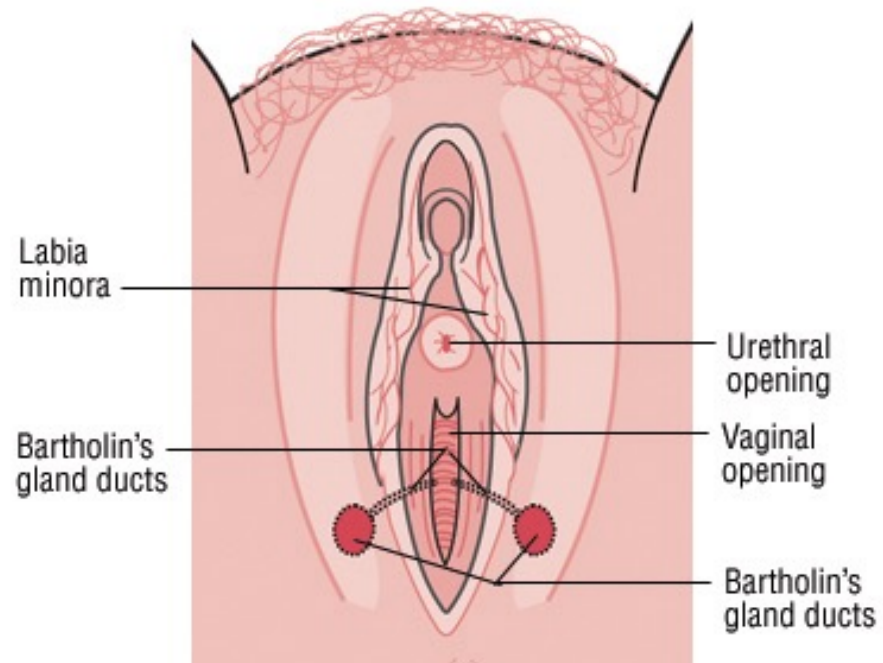
Clitoris



Various Types of Hymens



Bartholin gland



Breast

- The breasts are specialized **accessory glands of the skin** that secrete milk
- They are present in both sexes, in males and immature females, they are similar in structure rudimentary.
- The female breast overlies the **2nd–6th rib**.
- The breast is made up of **15–20 lobules** are **separated by fibrous septa running from the subcutaneous tissues** to the fascia of the chest wall (the ligaments of Cooper).
- Each **lobule drains by its lactiferous duct onto the nipple**, which is surrounded by the **pigmented areola**.

Breast

